

Prerequisites : theory of sinusoidal signals

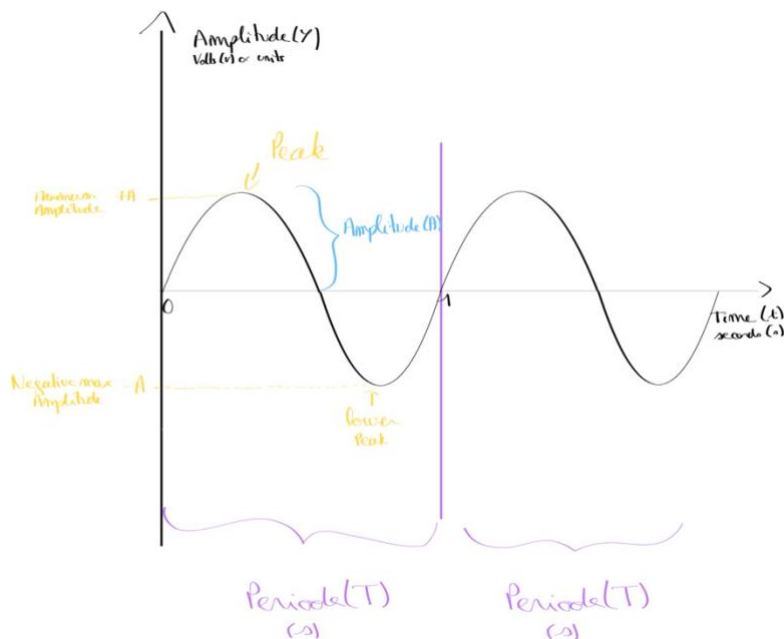
1. What is a sinusoid ?

A sine wave is a mathematical curve representing a regular oscillation described by a sine or cosine function

2. What (simple) physical phenomenon produces a sinusoid?

A sine wave is produced by the oscillation of a simple mechanical system, such as a weighted pendulum (which oscillates through a small angle). Name of the phenomenon : simple harmonic movement

3. Draw a diagram of a sine curve as a function of time. Indicate the name and units of the axes indicate the remarkable points



4. What is the equation of a sinusoid? write down the equation indicate the meaning of each term ; indicate the unit of each term ; indicate the unit of the whole equation

$$y(t) = A \sin(\omega t + \phi) + C \quad \text{or} \quad y(t) = A \cos(\omega t + \phi + C)$$

Y(t) — Instantaneous value

Meaning: The value of the signal or displacement at a specific instant t.

Unit: Depends on the physical quantity measured (e.g., meters m, volts V, etc.)

A — Peak amplitude

Meaning: The maximum variation or distance from the central baseline to the peak.

Unit: Same unit as the measured quantity (m, V, etc.)

Sin(...) — Sine trigonometric function

Meaning: The periodic oscillation function bounding the signal between -1 and +1.

Unit: Dimensionless.

Cos(...) — Cosine trigonometric function

Meaning: The periodic oscillation function bounding the signal between -1 and +1 (shifted ahead of the sine wave by « $\pi/2$ » radians or a quarter-period).

Unit: Dimensionless.

Omega — Angular frequency (pulsation)

Meaning: The rate of change of the oscillation's phase; related to frequency f by : $\omega = 2\pi f$
and to period T by : $\omega = \frac{2\pi}{T}$

Unit: Radians per second (rad/s).

t — Time

Meaning: Independent variable representing elapsed time.

Unit: Seconds (s).

φ — Initial phase (phase offset)

Meaning: The phase angle of the wave at time $t = 0$.

Unit: Radians (rad) or degrees ($^\circ$).

C — Vertical offset (DC offset)

Meaning: The central baseline or mean value around which the waveform oscillates.

Unit: Same unit as the measured quantity (m, V, etc.).

Unit of the Whole Equation

Unit: The overall unit is the unit of the physical quantity being measured (identical to the units of $y(t)$, A, and C).

5. Define (in plain English + give unit) and illustrate (on the diagram):

➤ **Amplitude (A)**

Definition: The maximum height or displacement of the wave measured from the central baseline to the highest point (crest) or lowest point (trough).

Unit: Depends on the signal (e.g., meters (m), volts (V), or dimensionless).

➤ **Period (T)**

Definition: The time it takes for the wave to complete one full cycle (from one peak (crest) to the next, or between two identical points).

Unit: Seconds (s).

➤ **Frequency (f)**

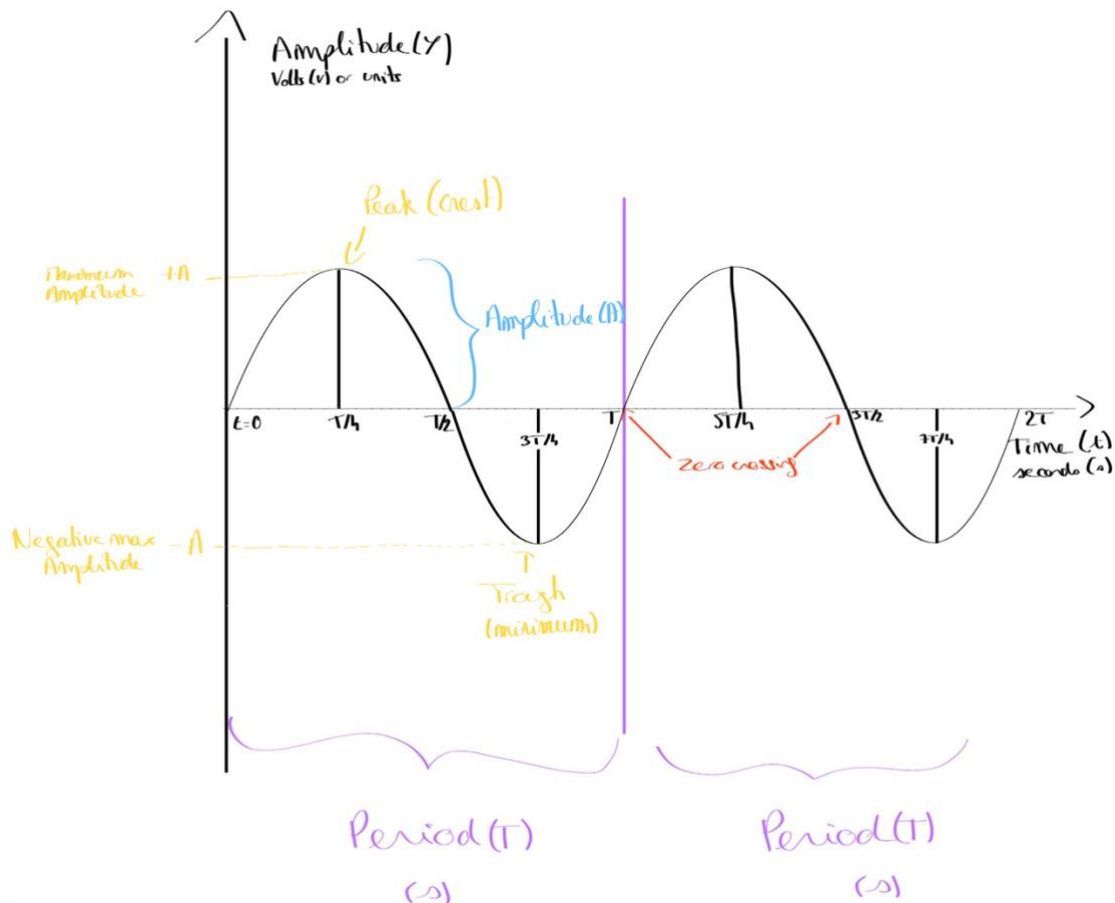
Definition: The number of complete cycles the wave performs in one second ($f = 1/T$).

Unit: Hertz (Hz) or inverse seconds (s^{-1}).

➤ **Pulsation / Angular Frequency (omega) (both mean the same thing)**

Definition: The rate of change of the wave's phase angle over time, describing how fast the oscillation rotates through a circle ($\omega = 2\pi f = \frac{2\pi}{T}$)

Unit: Radians per second (rad/s).



6. What is the difference between a sine and a cosine ? Write down the equation of a cosine compared to the equation of a sine draw a diagram of a cosine compared to the diagram of a sine

A sine wave and a cosine wave have the same waveform, the same amplitude and the same frequency; they differ only by a phase shift of $\pi/2$ radians (or 90°), which corresponds to a time shift of a quarter of a period ($T/4$).

At the origin ($t = 0$), the sine is 0 ($\sin(0) = 0$), whilst the cosine is already at its maximum ($\cos(0) = 1$). The cosine leads the sine by a quarter of a turn.

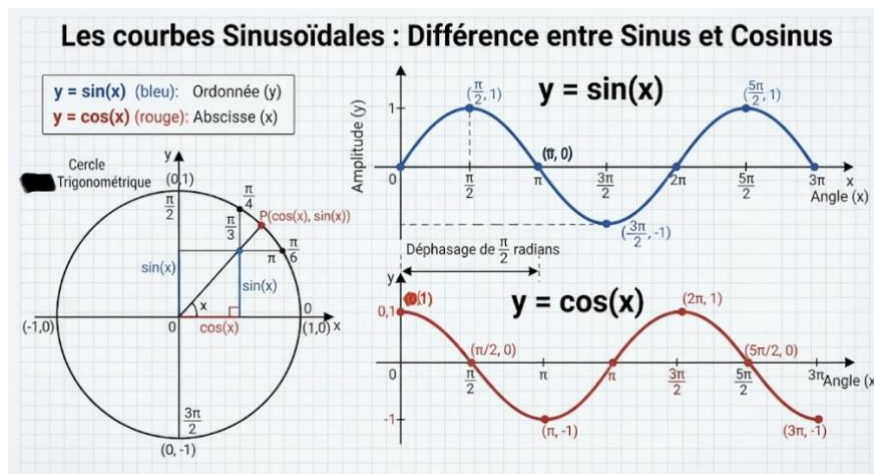
The cosine equation and a comparison with the sine equation

General cosine equation: $y(t) = A \cdot \cos(\omega t + \varphi) + C$

Mathematical comparison formula: $\cos(\omega t) = \sin\left(\omega t + \frac{\pi}{2}\right)$

(Or conversely : $\sin(\omega t) = \cos\left(\omega t - \frac{\pi}{2}\right)$)
The cosine curve corresponds to the sine curve shifted horizontally to the left by a duration of

$$\Delta t = \frac{T}{4}$$



7. Explain why a sinusoid is a periodic signal

A sinusoid repeats identically at regular, constant time intervals defined by its fundamental period.

Reference: Oppenheim, A. V., & Willsky, A. S. (1997). Signals and Systems (2nd ed.). Prentice Hall.

8. Explain why a sinusoid is a continuous signal

Its mathematical function is smoothly defined without any interruption across all real time.

Reference: Proakis, J. G., & Manolakis, D. G. (2007). Digital Signal Processing: Principles, Algorithms, and Applications (4th ed.). Pearson.

9. Explain why a sinusoid is a parametric signal, and with which parameters

Its waveform is completely defined by three key parameters: amplitude, frequency, and initial phase.

Reference: Smith, S. W. (1997). The Scientist and Engineer's Guide to Digital Signal Processing. California Technical Publishing.

10. URLs

1. <https://ocw.mit.edu/courses/res-6-007-signals-and-systems-spring-2011/> (MIT OpenCourseWare — Signals and Systems)
2. <https://dspguide.com> (Steven W. Smith — The Scientist and Engineer's Guide to Digital Signal Processing)
3. <https://mathworld.wolfram.com/Sine.html> (Weisstein, Eric W. Wolfram MathWorld — Sine)

Books / Scientific References:

1. Oppenheim, A. V., & Willsky, A. S. (1997). Signals and Systems (2nd ed.). Prentice Hall.
2. Proakis, J. G., & Manolakis, D. G. (2007). Digital Signal Processing: Principles, Algorithms, and Applications (4th ed.). Pearson.

3. Smith, S. W. (1997). The Scientist and Engineer's Guide to Digital Signal Processing.
California Technical Publishing

11. Time Spent on This Task

Time spent: Approximately 2h